

SAT UZBEKISTAN

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Module-1

1.

A right square pyramid has a total surface area of 19200 square inches and the combined surface area of the four lateral faces of the pyramid is 9984 square inches. What is the height of this pyramid, in inches?

2.

If $\left(\frac{a}{6}\right)x^2 + \left(\frac{b}{6}\right)x + \frac{c}{6} = -12(5x - 4)^2 + 6(5x - 3)^2$, find the sum of $a + b + c$.

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3.

$\sqrt{330 + 36\sqrt{6}} = r + \sqrt{p}$ What is the value of $r + p$?

4.

An equilateral triangle is inscribed in a circle. The perimeter is 780 and the circle has a radius of $a\sqrt{3}$. What is the value of a ?

5.

In triangles ABC and XYZ, $BC = YZ = 45$. Angles B and Y each have a measure of 60 degrees and

$\frac{AB}{XY} = \frac{BC}{YZ}$. Which additional information is needed to prove the two triangles congruent?

- A) $AB = XZ = 21$
- B) Angles A and X are each 90 degrees
- C) Angles Z and C are each 30 degrees
- D) No additional information is needed

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6.

A raised garden in the shape of a rectangular prism is filled with topsoil to a height of 24 inches. The garden has a width of 4 feet and a length of 20 feet. The total cost to fill the garden was \$231. What is the cost of the topsoil in dollars per cubic yard, rounded to the nearest cent? (1 foot = 12 inch) (1 yard = 3 feet)

7.

$f(x) = 2(x - a)(x - b)(x - c)$ where a , b , and c are distinct constants

When $a < x < b$, $f(x)$ is positive and point (r, s) is on the graph of $f(x)$. If $s = 6$, which of the following could be true?

- I. $r < a$
 - II. $b < r < c$
 - III. $r > c$
- A) I only
 B) I and III only
 C) III only
 D) II and III only

8.

Points $(1, 12)$ and $(c, 18)$ are on the line k . Points $(-10, c)$ and $(3, 30)$ are on the line m , where c is a constant. If the given lines are parallel, what is the sum of possible values of c ?

9.

Find the factors of $48x^4 + 200x^2 + 208$ in the form $ax^2 + b$, where a and b are integers. Then, find the minimum value of ab .

10.

If $x = \sqrt[2n]{2x^n + 35}$, where n is a positive integer constant, what is the value of x^n ?



11.

In the xy -plane, a parabola has a vertex $(7, 12)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants, which of the following could be the value of $a + b + c$?

- A) 10
B) 14
C) 18
D) 36
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12.

The function f is defined by $f(x) = 3^x$. The function g is an increasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points (a, j) and (b, k) , where $j < k$. When $g(x) > f(x)$, which of the following must be true?

- A) $x > k$
B) $x < j$
C) $x < j$ or $x > b$
D) $a < x < b$

13.

$$y = ax^2 + bx + c$$

The equation of a parabola is given, where a , b , and c are constants. In the xy -plane, the parabola has vertex $(5, -9)$ and has zero x -intercepts. Which of the following could be the value of $a - b - c$?

- A) -47 B) -5 C) 9 D) 23

14.

In the xy -plane, a parabola has vertex $(-5, -12)$ and has no x -intercepts. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants, which of the following could be the value of $a + b + c$?

- A) -19
- B) -12
- C) -10
- D) -5

15.

A line intersects 2 parallel lines creating 4 acute angles and 4 obtuse angles. One acute angle is $6x - 620$ and the sum of one acute angle and 3 obtuse angles is $-12x + w$. What is w ?

16.

1250 people went to the first webinar offered by a company. Of those 1250 people, 64% went to a second webinar. 32% of the people who went to the first & second webinars went to a third webinar. Of those who went to the first webinar but not the second webinar, 34% also went to the third webinar. How many people went to both the first & third webinars but not the second?

17.

The equation $8w^2 + 9w + c = 0$ has a solution of $\frac{-9 + \sqrt{497}}{16}$. What is the value of c ?

18.

The circle $2x^2 + 2y^2 + tx + ty - 60 = 0$ has a radius of $\sqrt{62}$ where t is a positive constant. What is the value of t ?

19.

The equation $N(m) = 61(Q)^{\frac{m}{4}}$ gives the number of cells in a petri dish m minutes after creation where Q is a positive constant. The number of cells increases by $p\%$ every 180 seconds. Which of the following expressions is equal to p ?

- A) $100(Q^{45} - 1)$
- B) $100\left(Q^{\frac{3}{4}} - 1\right)$
- C) $100(Q^3 - 1)$
- D) $100\left(Q^{\frac{3}{4}} + 1\right)$

20.

$6rx^2 - 24sx + 6r = 0$ where $r > 0$, $s < 0$, and $\frac{r}{s} > k$. Find the smallest possible value of k where the equation has 2 real solutions.

21.

There are 8 resistors in total in a circuit. The total resistance is equal to the sum of the individual resistance from each resistor. The circuit has a total resistance of 110 ohms. Three of the resistors have a total combined resistance of 70. Which of the inequalities represents the possible values for the individual resistance, x , of one of the 5 remaining resistors?

- A) $0 < x < 8$
- B) $0 < x < 40$
- C) $40 < x < 70$
- D) $0 < x < 110$

22.

Initially, there are the same number of prey & predators in a habitat. The number of prey increased by 1700 percent while the number of predators increased by 350 percent. The number of predators is what percent less than the number of prey after this increase?

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Module-2

1.

The function $h(x) = -\sqrt{x^2 + bx + c}$ passes through the points $(2,0)$ & $(0, -\sqrt{370})$ where b & c are constants. If $h(m) = 0$, what is the greatest possible value of m ?

2.

Arturo builds and sells two types of steel tables. The revenue Arturo generates from selling each square table is 2 times his cost to build a square table, and the revenue Arturo generates from selling each hexagonal table is 2.5 times his cost to build a hexagonal table. In one year, Arturo's total cost to build tables was \$ 42,120, and he generated a total revenue of \$ 101,250 from selling all the tables he built that year. For that year, how much greater is the revenue Arturo generated from selling hexagonal tables than the revenue he generated from selling square tables?

3.

The function $f(x) = \frac{x^2 + ax + b}{(2x + c)}$ where a , b , c are constants, does not touch the line $x = 3$ and $f(x) = 0$ for both $f(7)$ and $f(8)$. Find $a + b + c$.

4.

The circle $x^2 + x\sqrt{t} + y^2 - y - 69 = 0$ has a radius $= \sqrt{86}$. What is the value of t ?

- A) 71
- B) 73
- C) 35
- D) 38

5.

Two square pyramids have a height of 11 cm and 22 cm, respectively. The area of the base for each pyramid is 144 cm^2 . What is the positive difference in the surface areas between the two pyramids? Round to your answer tenth digit

6.

A circle has center G , and points M and N lie on the circle. Line segments MH and NH are tangent to the circle at points M and N , respectively. If the radius of the circle is 168 millimeters and the perimeter of quadrilateral $GMHN$ is 3,856 millimeters, what is the distance, in millimeters, between points G and H ?

- A) 168
- B) 1,752
- C) 1,760
- D) 1,768

7.

A circle in the xy -plane has its center at $(-4, -6)$. Line k is tangent to this circle at the point $(-7, -7)$. What is the slope of line k ?

- A. -3
- B. $-\frac{1}{3}$
- C. $\frac{1}{3}$
- D. 3

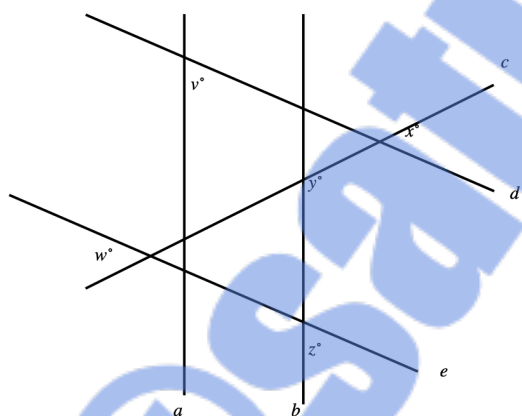
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8.

The quadratic function $y = ax^2 + bx + c$ intersects the x -axis at two points $(-9, 0)$ and $(4, 0)$. If $-5 < a < -1$, and a , b , and c are integer constants, which of the following is **not** a possible value of $a + b + c$?

- A) 150
- B) 120
- C) 90
- D) 60

9.



Note: Figure not drawn to scale.

In the figure, parallel lines a and b are intersected by lines c , d , and e . If $z = 67$, $y = 118$, and $v < z$, which statement about x and w must be true?

- A) $x < w$
- B) $x > w$
- C) $x = w$
- D) The relationship between x and w can not be determined.

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10.

The solutions $x^2 + 10x + 23 = 0$ are r and s , where $r < s$. The solutions $x^2 + 10x + 17 = 0$ are t and u , where $t < u$. The solutions to $x^2 + 20x + c = 0$, where c is a constant are $r + u$ and $s + t$. What is the value of c ?



11.

At the start of a day, a plant was 99 cm tall. It then grew $3 \frac{cm}{hr}$ for 4 hours. Then for the next 24 hours, it grew at a rate of $2 \frac{cm}{hr}$. Which equation below shows the height, H , in terms of x , number of hours since the start of the day, given that $4 \leq x \leq 28$?

- A) $H(x) = 99 + 2(x - 4)$
- B) $H(x) = 99 + 3 + 2(x - 4)$
- C) $H(x) = 99 + (4)(3) + 2(x - 4)$
- D) $H(x) = 99 + (4)(3) + 2x$

12.

A set designer is hammering nails along the top edge of a wall with a length of $3x$ feet, where x is an integer, that will be used to hang garland. The designer hammers the first nail at the left edge of the wall. The designer then hammers an additional nail every $\frac{3}{8}$ feet along the entire length of the wall, starting from the first nail and with the last nail being hammered at the right edge of the wall. Which equation best represents this situation, where y is the number of nails the designer hammers along the entire length of the wall?

- A) $y = \frac{3}{8}x$
- B) $y = \frac{3}{8}x + 1$
- C) $y = 8x$
- D) $y = 8x + 1$

13.

In isosceles triangle ABC , sides AB and AC are congruent. Point D divides side BC such that the length of BD is $\frac{4}{7}$ of the length of BC . Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD . If the length of BE is 12, where is the length of CF ?

- A) 9
- B) 21
- C) 48
- D) 84

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14.

A beekeeper's initial observation of the population of a certain bee colony was 1,100 bees. The beekeeper set a goal to increase the population to 2,600 bees. The beekeeper uses a model that predicts the population of this bee colony begins at 1,100 and increases by 120 bees per week in the first two weeks after the initial observation, and then increases by 180 bees per week until the beekeeper's goal is reached. According to this model, at the end of week w after the initial observation, where $w > 2$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

- A) $p(x) = 2,600 - 180w$
- B) $p(x) = 2,480 + 180w$
- C) $p(x) = 1,620 - 180w$
- D) $p(x) = -120 + 180w$

15.

Lines k and l are perpendicular and intersect at the point $(6, 3)$. Line k has a slope of m , which is a non-zero constant. Which of the following points does line l pass through?

- A) $(7, 3 - \frac{1}{m})$
- B) $(7, 3 + \frac{1}{m})$
- C) $(7, 3 - m)$
- D) $(7, 3 + m)$

16.

Triangle ABC has B as a right angle, $AC = 35$ and AB is 11 less than AC. A perpendicular line segment is drawn from Point B and intersects side AC at point D. What is the ratio of $\frac{BC}{BD}$?

17.

The expression kx represents the decrease of a positive force, x , that acts on an object by .21%. What is the value of k ?

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18.

Right triangle ABC is inscribed in a circle with angle B = 90 degrees, $BC > AB$ and AC is a diameter of the circle. BD is perpendicular to AC with point D on AC. $AC = 113$ and $BD = \sqrt{222}$. What is the ratio of $\frac{CD}{AD}$?

19.

There are 4 consecutive odd integers ordered from least to greatest with the first integer is expressed as x . The product of the first integer and 36 is at most 66 less than the sum of the first integer and the fourth integer. What is the maximum value that the first integer can be?

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A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week, the number of prey had increased by 2,700% of the number of prey at the start of the week, and the number of predators had increased by 180% of the number of predators at the start of the week. The number of prey at the end of the week was $p\%$ greater than the number of predators at the end of the week. What is the value of p ?

21.

$$y = 576^{(2x+2)}$$

The graph of the given equation in the xy -plane has a y -intercept of (r, s) . Which of the following equivalent equations displays the value of s as a constant, a coefficient, or the base?

A) $y = 331,776^{(x+1)}$

B) $y = 24^{(4x+4)}$

C) $y = \frac{1}{24}(24)^{(4x+5)}$

D) $y = \frac{1}{576}(576)^{(2x+3)}$

22.

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For the function f , for each increase in the value of x by c , where c is a positive constant, the value of $f(x)$ increases by a factor of 27. Which of the following equivalent forms of the function f displays $\frac{1}{c}$ as a coefficient of x ?

(A) $f(x) = 48(3)^{\frac{1}{c}x}$

(B) $f(x) = 48(3^{\frac{1}{c}})^x$

(C) $f(x) = 48(9)^{\frac{1}{c}x}$

(D) $f(x) = 48\left(27^{\frac{1}{c}x}\right)^{\frac{1}{2}}$